# Yuan Yuan

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## Education

**University of California, Los Angeles** Sep 2026 – May 2031 (Expected)
- Ph.D. in Computer Science

**University of Pennsylvania** Sep 2023 – May 2026
- M.S. Data Science & Computer and Information Science GPA: 3.88/4.0

**Vassar College** Sep 2017 – May 2021
- B.A., Major in Political Science, Correlate in Math GPA: 3.93/4.0

## Research Interests

Large Language Model, LLM Agent, Multiagent System, Multimodal LLM, LLM Post-Training & Reinforcement Learning, Synthetic Data

## Selected Research

**Analyzing Multi-Agent Inference Scaling: Communication and Shared Memory**
Work in Progress
- Conducting **quantitative and qualitative analyses** of CORAL-style inference with teams of up to 1024 agents on optimization tasks. Tracking scored submissions, agent trajectories, shared-memory activity, and peer interactions to study how team size affects coordination and task performance.
- Traced whether discoveries in one agent's trajectory were published, retrieved, and used by teammates; identified cases of unshared debugging insights and available information that went unused. Examined communication traces from Qwen and GPT-6 runs and compared agent harnesses to separate model behavior from differences in communication and context management.

**ImplicitPersona: Continuous Learning of Implicit User Personas via Reinforcement Fine-tuning for LLM Personalization**
Under Review of ARR January
- We built a **synthetic dataset** to evaluate and improve LLMs' ability to detect **implicit user preferences** in long-term conversations; using **Reinforcement Fine-Tuning (RFT)**, we design reward signals that encourage models to infer subtle, unspoken preferences and adapt responses accordingly. Our results show that RFT-trained models achieve stronger personalization and alignment with evolving user needs, advancing beyond memory evaluation toward more adaptive, context-aware AI assistants.
- Implemented a multi-modal RL pipeline to train the Qwen2 model on the dataset using VERL; implemented a DPO training pipeline using LLaMA-Factory; evaluated GPT, Qwen, LLaMA, and Claude models on the dataset.

*Bowen Jiang, **Yuan Yuan**, Zhuoqun Hao, Zhangchen Xu, Anvesh Rao Vijjini, Ziyi Liu, Radha Poovendran, Dan Roth, Camillo J. Taylor, Sihao Chen.*

**Know Me, Respond to Me: Benchmarking LLMs for Dynamic User Profiling and Personalized Responses at Scale**
COLM 2025 & Long abstract at the MASC-SLL 2025 [Oral]
- We propose **PersonaMem**, a benchmark built from over 180 simulated user–LLM interaction histories (each with up to 60 sessions over 15 tasks) that captures evolving user personas over time. Given an in-situ user query, models must choose responses aligned with the user's current profile state. We find that even leading LLMs struggle: they often fail to adapt to dynamic profile changes, achieving only ∼50% accuracy overall.
- Evaluated GPT, Qwen, Llama, and Claude models on the benchmark.

*Bowen Jiang*, Zhuoqun Hao*, Young-Min Cho, Bryan Li, **Yuan Yuan**, Sihao Chen, Lyle Ungar, Camillo J. Taylor*, Dan Roth*.*

## Research Experiences

**A Concise Agent is Less Expert: Revealing Side Effects of Using Style Features on Conversational Agents**
Under Review of ARR January

- We studied how defining personas with style features (e.g., "empathetic") shapes agent behavior and whether traits interact across dimensions. Our experiments test whether adjusting empathy meaningfully shifts responses and explore cross-trait effects, such as whether higher empathy decreases helpfulness or if conciseness and informativeness are inherently coupled. This work advances understanding of how personality traits influence conversational dynamics and how agents can better align with human perceptions of persona.
- Generated synthetic user-assistant conversations with GPT-4o-mini, Qwen3-8b and Llama3-8b (API calls and vllm); investigated side effect strength by calculating ELO ratings and win rates based on ELO Uncovered; investigated side effect significance with linear regression and t-test.

*Young-Min Cho*, **Yuan Yuan***, Sharath Chandra Guntuku, Lyle Ungar.*

**Cross-Cultural Advice: Comparative Analysis of LLM-Generated and Human Advice in Chinese and American Contexts**
Work in Progress

- We investigated **cross-cultural style variations** between Chinese and American advice-seeking platforms (Zhihu and Reddit) by developing a question-matching pipeline using **M3 embeddings** and **GPT-4o-mini** to align semantically similar queries across languages. This work advanced our understanding of LLMs' (GPT, Qwen) ability to replicate cultural nuances through **multilingual advice generation** and **culturally informed prompting**, by analyzing dimensions of style and human preferences.
- Generated LLM advices for collected questions; implementing a quantitative analysis of style features to compare LLM outputs against human-generated advice benchmarks; led human annotation to understand what LLM advices humans preferred.

*Sunny Rai, Jeffrey Cho, **Yuan Yuan**, Neil Sehgal, Sharath Chandra Guntuku, Lyle Ungar.*

**AgentWorld: A Multi-Agent Research Platform for Collaborative AI Systems**
COLM 2026

- Developed **AgentWorld**, an open-source research platform built on the Kaetram engine designed to study multi-agent coordination within a persistent 2D multiplayer environment. The project addresses the need for a high-fidelity sandbox where AI agents can move beyond simple logic to perform complex, long-horizon tasks such as crafting, trading, and combat. By providing a rich observation and action space via a unified API, AgentWorld enables the transition from isolated LLM benchmarking to **dynamic, multi-agent behavioral research**, offering significant implications for understanding how autonomous agents can collaborate or compete in resource-constrained environments.
- Designed and validated a series of **structured agent cooperation tasks** using a YAML-based Task Runner system; implemented specialized scenarios for smithing, mining, and exploration to benchmark LLM reasoning. Conducted **human annotation** of agent trajectories to evaluate the alignment between model actions and task objectives.

*Raphael Zhu, Jeffrey Cho, Yusen Zhang, Wenliang Zheng, **Yuan Yuan**, Jin Mo Yang, Chi Wang, Rui Zhang.*

**One Model, All Roles: Multi-Turn, Multi-Agent Self-Play Reinforcement Learning for Conversational Social Intelligence**
arXiv preprint arXiv:2602.03109, 2026

- We propose **OMAR (One Model, All Roles)**, a reinforcement learning framework that develops social intelligence through **multi-turn, multi-agent conversational self-play**, where a single model role-plays all conversation participants simultaneously to pursue long-term goals and learn complex social norms. We introduce **hierarchical advantage estimation** for training stability, and show on the SOTOPIA social environment and Werewolf strategy games that trained models exhibit emergent fine-grained social intelligence such as empathy, persuasion, and compromise seeking.

*Bowen Jiang, Taiwei Shi, Ryo Kamoi, **Yuan Yuan**, Camillo J. Taylor, Longqi Yang, Pei Zhou, Sihao Chen.*

**TurnaboutLLM: A Deductive Reasoning Benchmark from Detective Games**
EMNLP 2025 Main

- We introduced TurnaboutLLM, a framework + dataset built from detective games (Ace Attorney, Danganronpa) that challenges LLMs to detect contradictions between testimonies and evidence within lengthy narrative contexts. The task involves a large answer space and diverse reasoning types, making it especially difficult for

current models. We evaluate 12 state-of-the-art LLMs and identify shortcomings of popular techniques like chain-of-thought prompting and scaling context size.
- Collected, parsed, and transformed game data into a format for LLMs to interact with; annotated reasoning chains for the dataset; evaluated ChatGPT, Llama, and Qwen models on the benchmark and conducted qualitative and quantitative analysis.

***Yuan Yuan***, Muyu He*, Muhammad Adil Shahid, Jiani Huang, Ziyang Li, Li Zhang.*

**ControlText: Unlocking Controllable Fonts in Multilingual Text Rendering without Font Annotations**
EMNLP 2025 Findings
- We presented a new **text rendering and editing algorithm for diffusion models** that improves text generation and allow users to specify fonts to generate in images. Our approach preserves the font features by using a segmentation model and additional image filtering, alleviating the need of any ground-truth font labels.
- Implemented the text segmentation module based on the Rethinking Text Segmentation paper; evaluated our approach against AnyText for multi-language text generation in images.

*Bowen Jiang*, **Yuan Yuan***, Xinyi Bai, Zhuoqun Hao, Alyson Yin, Yaojie Hu, Wenyu Liao, Lyle Ungar, Camillo J. Taylor.*

**Multilingual Retrieval Augmented Generation for Culturally-Sensitive Tasks: A Benchmark for Cross-lingual Robustness**
ACL 2025 Main
- We introduced **BordIRLines**, a multilingual dataset of 251 territorial disputes (720 queries across 49 languages) to evaluate RAG on culturally and politically sensitive tasks. Experiments show that multilingual retrieval improves cross-lingual consistency and reduces geopolitical bias, while analyses reveal differences in citation patterns between high- and low-resource languages.
- Designed and monitored the human annotation procedure with over 100 participants in the study.

*Bryan Li, Fiona Luo*, Samar Haider*, Adwait Agashe*, Tammy Li†, Runqi Liu†, Muqing Miao†, Shriya Ramakrishnan†, **Yuan Yuan**†, Chris Callison-Burch.*

**Towards Rationality in Language and Multimodal Agents: A Survey**
NAACL 2025 Main
- We present a survey framing **rationality** in intelligent agents via four axioms: information grounding, logical consistency, invariance from irrelevant context, and preference orderability. We review how recent advances in multimodal and multi-agent systems—especially those combining LLMs with external tools, perception modules, and agent collaboration—help push toward more rational behavior, and we highlight open challenges in evaluation, architecture design, and bias mitigation.
- Researched and drafted the Logical Consistency sections.

*Bowen Jiang, Yangxinyu Xie, Xiaomeng Wang, **Yuan Yuan**, Zhuoqun Hao, Xinyi Bai, Weijie J. Su, Camillo J. Taylor, Tanwi Mallick.*

## Professional Experience

**AI Engineer**, Microsoft – Remote — Sep 2025 – Dec 2025
- Experimented with Supervised Finetuning (SFT) and Reinforcement Finetuning (RFT) to internalize LLM-as-a-Judge responses into model parameters, using **verl**.
- Fine-tuned Large Language Models with **RFT** on personalization benchmarks on Azure Cloud.
- Developed and debugged **inference pipelines** for open- and closed-source LLMs, monitoring performance on personalization benchmarks.
- Maintained an internal multi-party **LLM chat platform** built with **FastAPI** and **HTML** for real-time collaboration.
- **Deployed and integrated new models** into the platform, ensuring seamless user interaction and stability.
- Extended model capabilities by implementing **GPT-style function calling** for enhanced reasoning and tool use.

**Software Engineer Intern**, Meta – Seattle, US — May 2025 – Aug 2025
- Built and shipped features for the **Brand Safety Advertiser Experience Team**, collaborating with cross-functional product designers, PMs, and engineers.
- **Initiated and scoped the first AI Agent project** for the team; designed and implemented an internal **Oncall Domain Agent** using Hack and Context Engineering, improving oncall efficiency and developer productivity.
- Enhanced **millions** of advertisers' experience with Facebook Ads by **improving UIs in JavaScript** and **streamlining workflows**, significantly reducing friction in ad interactions.
- Developed an **ads matcher in Hack/PHP** to automatically detect and filter ads based on advertiser

requirements, increasing accuracy of ad delivery and covering over **20 million** ads on platforms.

**Open Source Maintainer**, AsyncAPI at Google Summer of Code – Remote May 2024 – Sep 2024

- Led the development of a **Slack bot** using **Django**, utilizing **AWS** for infrastructure management and **MongoDB** with vector search capabilities; the bot integrated message history, GitHub repository data, and website content to enhance search and retrieval functionalities.
- Led the **deprecation of the AsyncAPI Generator CLI**, transitioning to a more robust and integrated AsyncAPI CLI environment, enhancing platform functionality and streamlined development processes.
- Initiated and directed the **integration of templates into the generator repository**, improving developers' experience and code reusability.
- Used **Git** for version control and code reviews; resolved critical issues in **React** engine templates, improving functionality and reliability.

## Open-Source Contributions

**AsyncAPI (Google Summer of Code Maintainer):** Led the migration of the Generator CLI and integrated modular templates to enhance developer experience.

- **PRs:** #1251, #1234, #1213 **Issue:** #1249

**vLLM (Contributor):** Participated in architectural discussions regarding distributed inference reliability and multi-node synchronization.

- **Issue:** #33034 (Proposed a benchmark-driven auto-tuning solution to replace hard-coded Mamba kernel configurations with a dynamic JSON-based loader)

## Honors

PHI BETA KAPPA Membership
The Julia Flitner Lamb Prize at Vassar College
Ann Cornelisen Fund for Post-Graduate Fellowships

## TECHNICAL SKILLS & FRAMEWORKS

Java, Python, R, C++, Spring Boot, Django, AWS EC2, AWS S3, AWS Kinesis, AWS RDS, AWS Aurora, AWS Elasticache, Redis, Apache Kafka, MySQL, Postgres, DynamoDB, MongoDB, Docker, Docker-compose, Kubernetes, Jenkins, ArgoCD, gRPC, HTTP, WebSocket